
Introduction to Linear Programming

Linear programming (LP) is the task of maximizing or minimizing a linear function over the set of solutions to a finite system of linear inequalities. Despite the apparent simplicity of the model, an enormous variety of problems — from production planning and network flows to scheduling, game theory and machine learning — can be expressed in exactly this form. The subject took its modern shape in the 1940s: Kantorovich [Kan60] and Koopmans developed LP models for economic planning, and Dantzig [Dan51, Dan63] introduced the general model together with the *simplex method*, still one of the dominant algorithms today. Duality, the subject of the last part of this lecture, emerged from a 1947 conversation between Dantzig and von Neumann and was first published by Gale, Kuhn and Tucker [GKT51]; the elimination method for linear inequalities that we will use to prove it is a century older still, going back to Fourier [Fou27]. Each of these threads runs through today's lecture.

This first lecture introduces the foundations. We introduce the LP model and how to visualize it (Section 1), develop the basic geometry of polyhedra — convexity, faces, vertices and rays, culminating in Minkowski's theorem — and deduce the possible solution statuses of an LP (Section 2), and finally prove the fundamental duality theorems, which allow us to *certify* optimality (Section 3).

1 The Linear Programming Model

A *linear program* is an optimization problem of the form

$$\begin{aligned} & \max \text{ (or min) } c^\top x \\ & \text{subject to } a_i^\top x \leq b_i, \quad i \in [m], \end{aligned} \tag{1}$$

where $x \in \mathbb{R}^n$ is the vector of *decision variables* and the data consists of the *objective* $c \in \mathbb{R}^n$ and the *constraints* $(a_i, b_i) \in \mathbb{R}^n \times \mathbb{R}$, $i \in [m] := \{1, \dots, m\}$. Constraints of the form $a_i^\top x \geq b_i$ or $a_i^\top x = b_i$ are also allowed: a $\geq$ constraint is a $\leq$ constraint after multiplying by -1 , and an equality is a pair of opposing inequalities. Similarly, minimizing $c^\top x$ is the same as maximizing $(-c)^\top x$. So all variants of (1) can be converted into one another, and we are free to work with whichever form is most convenient. Stacking the constraints into a matrix $A \in \mathbb{R}^{m \times n}$ with rows $a_1^\top, \dots, a_m^\top$, we arrive at the compact form we will use most often:

$$\max \{c^\top x : Ax \leq b\}, \quad b = (b_1, \dots, b_m)^\top,$$

where the inequality $Ax \leq b$ is interpreted coordinate-wise.

A point x satisfying all the constraints is called *feasible*, and the set $P := \{x \in \mathbb{R}^n : Ax \leq b\}$ of all such points is the *feasible region*. The *optimal value* is $\text{OPT} := \sup\{c^\top x : x \in P\}$, and a feasible x^* with $c^\top x^* = \text{OPT}$ is an *optimal solution*. Two degenerate situations can occur: the LP may be *infeasible* ($P = \emptyset$), in which case we set $\text{OPT} = -\infty$ by convention (for a maximization problem), or *unbounded*, meaning that $c^\top x$ takes arbitrarily large values over P , in which case $\text{OPT} = +\infty$. A main goal of this lecture is to understand these possibilities and how to recognize them.

1.1 Two examples

Production planning. A small workshop produces chairs and tables. Each chair yields a profit of 200 euros and each table 300 euros. Producing a chair consumes 1 unit of wood and 1 hour of

labor, while a table consumes 1 unit of wood and 3 hours of labor. With 4 units of wood and 6 hours of labor available, how should the workshop maximize profit? Writing $x_1, x_2 \geq 0$ for the number of chairs and tables produced (allowing fractional furniture for now), and measuring profit in hundreds of euros, we obtain the LP

$$\begin{aligned} \max \quad & 2x_1 + 3x_2 \\ \text{s.t.} \quad & x_1 + x_2 \leq 4 \quad (\text{wood}) \\ & x_1 + 3x_2 \leq 6 \quad (\text{labor}) \\ & x_1, x_2 \geq 0. \end{aligned} \tag{2}$$

The optimal solution turns out to be $(x_1, x_2) = (3, 1)$ with profit 9. It is easy to *check* that this point is feasible, but how can one certify that *no* feasible point does better, without checking them all? Answering this question is the subject of duality (Section 3).

Maximum flow. LPs can encode genuinely combinatorial problems. Let $G = (V, E)$ be a directed graph with source s , sink t , and capacities $u_e \geq 0, e \in E$. A *flow* assigns a value $f_e \in [0, u_e]$ to each edge so that flow is conserved at every internal node, and the goal is to push as much flow as possible from s to t :

$$\max \sum_{e \text{ out of } s} f_e \quad \text{s.t.} \quad \underbrace{\sum_{e \text{ into } v} f_e = \sum_{e \text{ out of } v} f_e}_{\text{flow conservation}}, \quad \forall v \in V \setminus \{s, t\}, \quad 0 \leq f_e \leq u_e.$$

This is an LP with $|E|$ variables and $O(|V| + |E|)$ constraints.

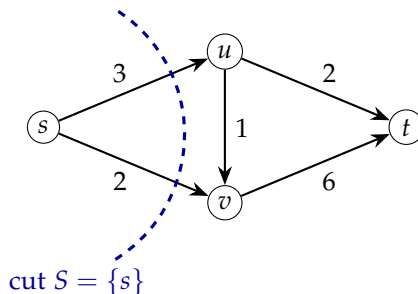


Figure 1: A maximum flow instance; edge labels are capacities. The maximum flow value is 5, attained by sending 3 units along $s \rightarrow u$ (of which 2 continue to t and 1 to v) and 2 units along $s \rightarrow v$. The dashed curve marks the cut $S = \{s\}$: every unit of flow must cross it, so no flow value can exceed its capacity $3 + 2 = 5$ — certifying optimality. Note that the arc $v \rightarrow t$ is not tight at this maximum flow.

1.2 The geometric picture

Each constraint $a_i^\top x \leq b_i$ (with $a_i \neq 0$) cuts space into two pieces: the *hyperplane* $\{x : a_i^\top x = b_i\}$, a line in $\mathbb{R}^2$ or a plane in $\mathbb{R}^3$, is the boundary, and the constraint keeps the *halfspace* on the side of the boundary that the normal vector a_i points *away* from. The feasible region is the intersection of these m halfspaces — a convex region with flat sides called a *polyhedron*.

The objective also has a geometric meaning: the level sets $\{x : c^\top x = \gamma\}$ form a family of parallel hyperplanes, sweeping in direction c as γ increases. To solve the LP visually, slide the

level set in the direction c until it is about to leave the feasible region: the last points of contact are the optimal solutions. Figure 2 shows this for the production LP (2).

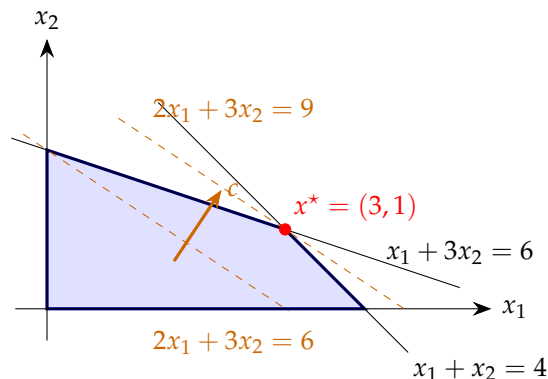


Figure 2: The production LP (2). The feasible region is the intersection of four halfspaces; sliding the objective level line in direction $c = (2, 3)$, the last contact point is the vertex $x^* = (3, 1)$, so $\text{OPT} = 9$.

Two features of Figure 2 are worth recording, since they hold in full generality and underlie everything that follows. First, the feasible region has no “holes”. Second, the optimum is attained at a *corner* (vertex) of the region. We will prove these properties formally in the next section.

2 Structure of Linear Programs

DEFINITION 1 (POLYHEDRON) A polyhedron is a set of the form $P = \{x \in \mathbb{R}^n : Ax \leq b\}$ for some $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, i.e., a finite intersection of halfspaces. A bounded polyhedron is called a polytope.

Polyhedra are exactly the feasible regions of linear programs. In this section we develop their basic structure and use it to classify the possible outcomes of an LP.

2.1 Convexity

Geometrically, a set is convex if it contains the line segment between any two of its points. Algebraically, the segment between x and y is the set of *convex combinations* $\lambda x + (1 - \lambda)y$, $\lambda \in [0, 1]$: at $\lambda = 1$ we obtain x , at $\lambda = 0$ we obtain y , and in between we interpolate linearly.

DEFINITION 2 (CONVEX SET) A set $C \subseteq \mathbb{R}^n$ is convex if for all $x, y \in C$ and $\lambda \in [0, 1]$, we have $\lambda x + (1 - \lambda)y \in C$.

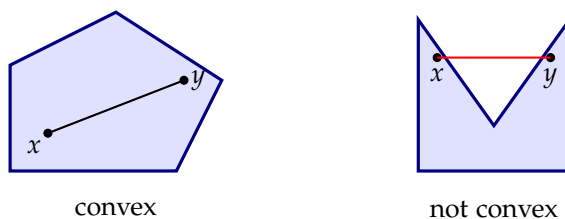


Figure 3: Convexity: the segment $[x, y]$ must stay inside the set.

PROPOSITION 3 Every polyhedron $P = \{x : \mathbf{A}x \leq b\}$ is convex.

PROOF: Let $x, y \in P$ and $\lambda \in [0, 1]$. Then $\mathbf{A}(\lambda x + (1 - \lambda)y) = \lambda \mathbf{A}x + (1 - \lambda)\mathbf{A}y \leq \lambda b + (1 - \lambda)b = b$, where we used $\lambda, 1 - \lambda \geq 0$ to preserve the inequalities. $\square$

Convexity has an important algorithmic consequence: to be globally optimal it suffices to be locally optimal, so an algorithm that greedily improves the objective cannot terminate at a suboptimal local optimum. Call $x \in P$ a *local maximum* of $c^\top x$ over P if there is $\varepsilon > 0$ such that $c^\top x \geq c^\top z$ for every $z \in P$ with $\|z - x\| \leq \varepsilon$.

PROPOSITION 4 (LOCAL IMPLIES GLOBAL) If x is a local maximum of $c^\top x$ over a convex set P , then x is a global maximum.

PROOF: Suppose some $y \in P$ has $c^\top y > c^\top x$. By convexity, the point $z_\lambda := \lambda y + (1 - \lambda)x$ is in P for all $\lambda \in [0, 1]$, and $c^\top z_\lambda = c^\top x + \lambda(c^\top y - c^\top x) > c^\top x$ for $\lambda > 0$. As $\lambda \rightarrow 0$, $z_\lambda \rightarrow x$, so every neighborhood of x contains feasible points of strictly larger value, contradicting local maximality. $\square$

2.2 Facial structure of polyhedra

Looking at a 3-dimensional polytope such as the cube, its boundary decomposes into corners, edges, and flat sides. The notion of a *face* captures all of these at once. Intuitively, a face is a subset of P that is closed under “de-averaging”: if a point of the face is a strict average of two points of P , both points must already lie in the face. One may think of a face as a set of optimal solutions: if x maximizes a linear function and x is the average of two feasible y, z , then y and z must be optimal too (averaging cannot hide a better point on one side without a worse one on the other).

DEFINITION 5 (FACE) A face of a polyhedron P is a convex subset $F \subseteq P$ such that whenever $\lambda x + (1 - \lambda)y \in F$ for some $x, y \in P$ and $\lambda \in (0, 1)$, we have $x, y \in F$. The dimension of F is the dimension of the smallest affine subspace $\text{aff}(F)$ containing it. Faces of dimension 0, 1, and $\dim(P) - 1$ are called vertices, edges, and facets respectively. Note that P itself and $\emptyset$ are always faces of P .

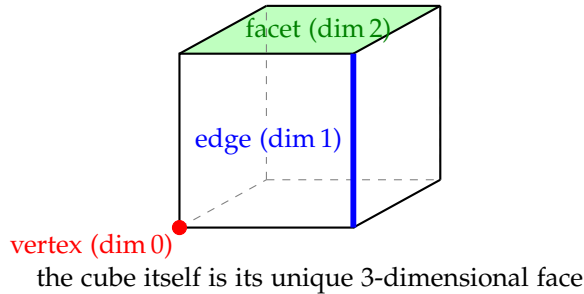


Figure 4: Faces of the cube $[0, 1]^3$, one highlighted per dimension.

The definition of a face is purely geometric — no inequalities appear in it. The fundamental theorem below gives the matching algebraic description: the nonempty faces of P are obtained by setting some of the defining inequalities to equality. For $I \subseteq [m]$, we abbreviate $\mathbf{A}_I := \mathbf{A}_{I, \bullet}$, the row

submatrix indexed by I — subscripts on $\mathbf{A}$ select rows throughout this lecture — and write b_I for the corresponding entries of b . Define

$$P_I := \{x \in P : \mathbf{A}_I x = b_I\}.$$

THEOREM 6 (FACES = TIGHT CONSTRAINT SETS) *Let $P = \{x : \mathbf{A}x \leq b\}$ be a polyhedron in $\mathbb{R}^n$ and let $F \subseteq P$ be nonempty. Then F is a face of P if and only if $F = P_I$ for some $I \subseteq [m]$. Moreover, taking I to be the set of constraints that are tight everywhere on F ,*

$$I := \{i \in [m] : a_i^\top x = b_i \ \forall x \in F\}, \quad \text{aff}(F) = \{x \in \mathbb{R}^n : \mathbf{A}_I x = b_I\},$$

and hence $\dim F = n - \text{rank}(\mathbf{A}_I)$.

PROOF: ($\Leftarrow$) Let $F = P_I$, which is convex as the intersection of the convex sets P and an affine subspace. Suppose $z = \lambda x + (1 - \lambda)y \in P_I$ with $x, y \in P$, $\lambda \in (0, 1)$. For each $i \in I$,

$$b_i = a_i^\top z = \lambda a_i^\top x + (1 - \lambda) a_i^\top y \leq \lambda b_i + (1 - \lambda)b_i = b_i,$$

so equality holds throughout, which forces $a_i^\top x = a_i^\top y = b_i$ (a strict average of two numbers $\leq b_i$ equals b_i only if both equal b_i). Hence $x, y \in P_I$, as needed.

($\Rightarrow$) Let F be a nonempty face and let I be as in the theorem statement, the set of constraints tight *everywhere* on F ; clearly $F \subseteq P_I$. Pick a point $x^0 \in F$. We first build a point $\bar{x} \in F$ at which every constraint outside I is *slack*: if $I \neq [m]$ (there is at least one non-tight constraint), for each $i \notin I$ pick $x^i \in F$ with $a_i^\top x^i < b_i$ (which exists by definition of I), and let $\bar{x} := \frac{1}{m-|I|} \sum_{i \in [m] \setminus I} x^i$ be the average of these points. By convexity of F we have $\bar{x} \in F$, and $a_i^\top \bar{x} < b_i$ for all $i \notin I$, since the average includes at least one strictly slack term. If $I = [m]$, take $\bar{x} := x^0$.

Now let $y \in P_I$ be arbitrary; we claim $y \in F$, establishing $P_I \subseteq F$. For $\delta > 0$ consider the point $w := \bar{x} + \delta(\bar{x} - y)$ obtained by walking from y through $\bar{x}$ and slightly beyond, so that $\bar{x} = \frac{1}{1+\delta}w + \frac{\delta}{1+\delta}y$ is a strict average of w and y . For $i \in I$ we have $a_i^\top w = b_i + \delta(b_i - b_i) = b_i$, while for $i \notin I$ the constraint is slack at $\bar{x}$, so $a_i^\top w \leq b_i$ for δ small enough. Thus $w \in P$ for small $\delta > 0$, and since $\bar{x} \in F$ is a strict average of $w, y \in P$, the face property gives $y \in F$.

For the affine hull, note the argument just given shows more: for *any* y with $\mathbf{A}_I y = b_I$ (feasible or not), the point $\bar{x} + \varepsilon(y - \bar{x})$ lies in $P_I = F$ for $\varepsilon > 0$ small (constraints in I remain tight, slack constraints remain slack). Hence the line through $\bar{x}$ and y meets F in a segment, so $y \in \text{aff}(F)$. This shows $\{x : \mathbf{A}_I x = b_I\} \subseteq \text{aff}(F)$, and the reverse inclusion is clear since $F \subseteq \{x : \mathbf{A}_I x = b_I\}$, an affine subspace. $\square$

An important consequence is of the above theorem is that a polyhedron, though it contains infinitely many points, has only *finitely many* faces. Facial structure provides a combinatorial way to understand polyhedra, a theme will explore further in the next lecture.

COROLLARY 7 (FACE COUNTING) *A polyhedron $P = \{x : \mathbf{A}x \leq b\}$ with m constraints has at most $\binom{m}{n-k}$ faces of dimension k . In particular, P has at most $\binom{m}{n}$ vertices, and every vertex is the unique solution of the linear system $\mathbf{A}_J x = b_J$ for some $J \subseteq [m]$ with $|J| = n$ and $\mathbf{A}_J$ invertible.*

PROOF: Let F be a face of dimension k with tight set I as in Theorem 6, so $\text{rank}(\mathbf{A}_I) = n - k$. Choose $J \subseteq I$ with $|J| = n - k$ indexing a basis of the row space of $\mathbf{A}_I$. Then $\{x : \mathbf{A}_I x = b_I\} \subseteq \{x : \mathbf{A}_J x = b_J\}$ are nonempty affine subspaces of the same dimension k , hence equal. Consequently

$$F = P \cap \text{aff}(F) = P \cap \{x : \mathbf{A}_J x = b_J\} = P_J,$$

so F is determined by the $(n - k)$ -element set J , and the number of k -faces is at most the number of such sets. For $k = 0$: $\text{aff}(F) = \{x : \mathbf{A}_J x = b_J\}$ is a single point with $|J| = n$, so $\mathbf{A}_J$ is invertible and the vertex is $\mathbf{A}_J^{-1} b_J$. $\square$

From the outer to the inner description. So far we have described polyhedra from the *outside*, as intersections of halfspaces, and studied the face structure this induces. The culminating result of this section is a description from the *inside*: every polyhedron is generated by finitely many points and directions. For finite sets $V, R \subseteq \mathbb{R}^n$, define

$$\text{conv}(V) := \left\{ \sum_{v \in V} \lambda_v v : \lambda \geq \mathbf{0}, \sum_{v \in V} \lambda_v = 1 \right\}, \quad \text{cone}(R) := \left\{ \sum_{r \in R} \mu_r r : \mu \geq \mathbf{0} \right\},$$

the *convex hull* of V and the *cone generated by* R , with the conventions $\text{conv}(\emptyset) = \emptyset$ and $\text{cone}(\emptyset) = \{\mathbf{0}\}$. We will combine the two using the *Minkowski sum* of sets, $A + B := \{a + b : a \in A, b \in B\}$.

In general, a set $C \subseteq \mathbb{R}^n$ is called a *cone* if it is convex and is closed under nonnegative scaling: $\lambda x \in C$ for all $x \in C$ and $\lambda \geq 0$.

The following proposition justifies the name convex hull and generated cone, and states that the Minkowski sum operation preserves convexity.

PROPOSITION 8 *The following convexity properties hold:*

1. $\text{conv}(V)$ is convex and $\text{cone}(R)$ is a cone for $V, R \in \mathbb{R}^n$ finite.
2. $A + B$ is convex if $A, B \subseteq \mathbb{R}^n$ are convex.

We will the proof as an exercise.

Exercise 1 Prove Proposition 8.

A *ray* is a set of the form $x + [0, \infty)d := \{x + td : t \geq 0\}$ with direction $d \neq \mathbf{0}$; when $x = \mathbf{0}$ we simply write $[0, \infty)d$, a *half-line* from the origin. Note that a ray starting at the origin is a cone. The recession cone, defined next, is a further key example of a cone.

DEFINITION 9 (RECESSION CONE) *The recession cone of $P = \{x \in \mathbb{R}^n : \mathbf{A}x \leq b\} \neq \emptyset$ is*

$$\text{rec}(P) := \{d \in \mathbb{R}^n : \mathbf{A}d \leq \mathbf{0}\},$$

the set of directions in which P recedes: if $x \in P$ and $d \in \text{rec}(P)$, then $\mathbf{A}(x + td) = \mathbf{A}x + t\mathbf{A}d \leq b$ for all $t \geq 0$, so the whole ray $x + td$ stays in P ; conversely, the direction of any ray contained in P lies in $\text{rec}(P)$ (a routine verification: if some $a_i^\top d > 0$, the constraint i is eventually violated along the ray). In particular, $\text{rec}(P)$ depends only on the set P , not on its description.

DEFINITION 10 (POINTED POLYHEDRON) *A nonempty polyhedron P is pointed if its recession cone contains no line, that is, if $d, -d \in \text{rec}(P)$ implies $d = \mathbf{0}$.*

When $\text{rank}(\mathbf{A}) = n$, the polyhedron P is automatically pointed: $d, -d \in \text{rec}(P)$ gives both $\mathbf{A}d \leq \mathbf{0}$ and $-\mathbf{A}d \leq \mathbf{0}$, so $\mathbf{A}d = \mathbf{0}$ and hence $d = \mathbf{0}$; the same argument shows that $\text{rec}(P)$, a polyhedron in its own right, is pointed as well. For the remainder of this subsection we assume $\text{rank}(\mathbf{A}) = n$; the results below use this standing hypothesis. Note also that P is bounded exactly when $\text{rec}(P) = \{\mathbf{0}\}$: a nonzero recession direction yields an unbounded ray, while the converse direction follows from Theorem 13 below, which in this case represents P as $\text{conv}(V(P))$, a bounded set.

OBSERVATION 11 (EDGES OF $\text{rec}(P)$ ARE HALF-LINES FROM $\mathbf{0}$) By Theorem 6 applied to the system $\mathbf{A}d \leq \mathbf{0}$, every nonempty face of $\text{rec}(P)$ has the form $\{d : \mathbf{A}d \leq \mathbf{0}, \mathbf{A}_I d = \mathbf{0}\}$, and is therefore closed under nonnegative scaling; in particular, every nonempty face contains $\mathbf{0}$, and the unique vertex of $\text{rec}(P)$ is $\mathbf{0}$ (a single point closed under nonnegative scaling must be the origin). Now let E be an edge (one-dimensional face) of $\text{rec}(P)$ and pick $r \in E \setminus \{\mathbf{0}\}$. Then $\text{aff}(E) = \text{span}(r)$, and E contains $[0, \infty) \cdot r$ by scaling but no negative multiple of r , as $\text{rec}(P)$ is pointed. Hence

$$E = [0, \infty) \cdot r :$$

every edge of $\text{rec}(P)$ is a half-line emanating from the unique vertex $\mathbf{0}$, unbounded in its generating direction r .

DEFINITION 12 (GENERATORS $V(P)$ AND $R(P)$) Let $V(P)$ denote the set of vertices of P , and let $R(P)$ contain one generator per edge of $\text{rec}(P)$: by Observation 11, each edge is a half-line $[0, \infty) \cdot r$, and $R(P)$ picks one such $r \neq \mathbf{0}$ for each (the choice of scaling will not matter). Edges of a pointed cone are also called its extreme rays — for pointed cones the two terms mean exactly the same thing, and we use them interchangeably. By Corollary 7, both sets are finite: $|V(P)| \leq \binom{m}{n}$ and $|R(P)| \leq \binom{m}{n-1}$. When P is bounded, $\text{rec}(P) = \{\mathbf{0}\}$ has no edges, so $R(P) = \emptyset$ and $\text{cone}(R(P)) = \{\mathbf{0}\}$: the conventions above handle the bounded case without special treatment.

THEOREM 13 (MINKOWSKI [MIN96]) Let $P = \{x : \mathbf{A}x \leq b\}$ be nonempty with $\text{rank}(\mathbf{A}) = n$. Then

$$P = \text{conv}(V(P)) + \text{cone}(R(P)).$$

In particular, $V(P) \neq \emptyset$.

REMARK 1 (THE MINKOWSKI–WEYL THEOREM) The converse of Theorem 13 is also true: every set of the form $\text{conv}(V) + \text{cone}(R)$ with V, R finite is a polyhedron (Weyl’s theorem). The converse direction will follow as a consequence of Fourier–Motzkin elimination (Corollary 21), which we cover in Section 3.2. The two directions say that the outer (halfspace) and inner (generator) descriptions capture *exactly* the same class of sets.

REMARK 2 As stated, Minkowski’s theorem does not hold when $\text{rank}(\mathbf{A}) < n$; equivalently, when $L := \ker(\mathbf{A}) \neq \emptyset$. The decomposition fails in this case because both $V(P)$ and $R(P)$ end up being empty (for $V(P)$ this is because all non-empty faces have dimension at least $\dim(L) \geq 1$). A correct Minkowski-like decomposition in this setting is given instead by

$$P = \text{conv}(V(P')) + \text{cone}(R(P')) + L,$$

where $P' := P \cap L^\perp$ and $L^\perp$ is the orthogonal complement of L . To verify this, one must check that $P = P' + L$ and that P' is a pointed polyhedron (so Theorem 13 applies to P').

We will require the following lemma, which provides a simple way to isolate directions that avoid the recession cone $\mathbf{A}x \leq \mathbf{0}$.

LEMMA 14 (BLOCKING LEMMA) For $\mathbf{A} := (a_1, \dots, a_m)^\top \in \mathbb{R}^{m \times n}$ with $\text{rank}(\mathbf{A}) = n$ and let $w_{\mathbf{A}} := \sum_{i=1}^m a_i$. Let $d \neq \mathbf{0}$ satisfy $w_{\mathbf{A}}^\top d = 0$. Then, we can row $a_j^\top d > 0$, which we say blocks d , for some $j \in [m]$; and likewise in direction $-d$.

PROOF: If no row blocks direction d , then $\mathbf{A}d \leq \mathbf{0}$, so $0 = w^\top d = \sum_{i=1}^m a_i^\top d$ is a vanishing sum of nonpositive terms. Hence every term is zero, so $\mathbf{A}d = \mathbf{0}$, contradicting that $\mathbf{A}$ has full column rank and $d \neq \mathbf{0}$. The same argument applies to $-d$. $\square$

We will use the following geometric consequence: for any point $x \in P$ and direction d as above, the ray $x + [0, \infty)d$ is eventually “blocked” by some constraint of P , that is, its intersection with P is bounded. To see this, simply note that $a_j^\top(x + td)$ will exceed the right hand side b_j if a_j blocks d and $t \geq 0$ is large enough.

PROOF OF THEOREM 13: ($\supseteq$) A convex combination of vertices lies in P since convex sets are closed under finite convex combinations (use induction on the number of terms in the combination), and adding an element of $\text{cone}(R(P)) \subseteq \text{rec}(P)$ keeps us inside P by Definition 9.

($\subseteq$) Let $x \in P$, let $I = I(x)$ be its set of tight constraints, and let $k := \text{rank}(\mathbf{A}_I)$. We show $x \in \text{conv}(V(P)) + \text{cone}(R(P))$ by induction on $n - k$.

Case $k = n$. Then $\{x\} = P_I$ is a face of dimension 0 by Theorem 6, i.e., $x \in V(P)$.

Case $k \leq n - 2$. Here $\ker(\mathbf{A}_I)$ has dimension at least 2, and thus by dimension counting there exists a non-zero $d \neq \mathbf{0}$ with $\mathbf{A}_I d = \mathbf{0}$ and $w_A^\top d = 0$ (stacking $w_A^\top$ on top of $\mathbf{A}$ can only reduce the dimension of the kernel by 1). By Lemma 14, the rays emanating from x in the directions d and $-d$ are both blocked; let $t^\pm > 0$ be the largest feasible step sizes (strictly positive: every blocking constraint j has $a_j^\top d \neq 0$, hence $j \notin I$, hence is slack at x), and set $x^\pm := x \pm t^\pm d$. Each of $x^\pm$ keeps the tight constraints I (as $\mathbf{A}_I d = \mathbf{0}$) and gains a new tight constraint j with $a_j^\top d \neq 0$; such a row is linearly independent of the rows of $\mathbf{A}_I$, so the tight rank of $x^\pm$ is at least $k + 1$. By induction, x^- and x^+ lie in $\text{conv}(V(P)) + \text{cone}(R(P))$, and hence so does the convex combination

$$x = \frac{t^+}{t^+ + t^-} x^- + \frac{t^-}{t^+ + t^-} x^+$$

(the set $\text{conv}(V(P)) + \text{cone}(R(P))$ is convex, being a sum of convex sets).

Case $k = n - 1$. Now $\ker(\mathbf{A}_I) = \text{span}(d)$ for a single direction $d \neq \mathbf{0}$, and we can no longer insist on $w^\top d = 0$. Still, at least one of $\pm d$ is blocked: being blocked in neither direction would give $\mathbf{A}d \leq \mathbf{0}$ and $\mathbf{A}(-d) \leq \mathbf{0}$, i.e., $d = \mathbf{0}$ since $\text{rank}(\mathbf{A}) = n$. If both are blocked, then exactly as in the previous case, x is a convex combination of x^- and x^+ , which now have tight rank n : two vertices. If only, say, direction $-d$ is blocked, then $\mathbf{A}d \leq \mathbf{0}$, so $d \in \text{rec}(P)$ and $x = x^- + t^- d$ with x^- a vertex. Moreover, $F := \{e \in \text{rec}(P) : \mathbf{A}_I e = \mathbf{0}\}$ is a face of $\text{rec}(P)$ (as in Observation 11) that contains d and is contained in $\ker(\mathbf{A}_I) = \text{span}(d)$, hence an edge of $\text{rec}(P)$; by Observation 11, $F = [0, \infty) \cdot d$. Thus d is a positive multiple of the generator of F chosen in $R(P)$, and $x \in V(P) + \text{cone}(R(P))$. $\square$

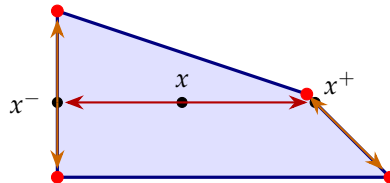


Figure 5: The two-sided walk behind Theorem 13: x is a strict convex combination of $x^\pm$, each lying on a face of smaller dimension; recursing reaches the vertices (red). If P is unbounded, a final-level walk may instead run off to infinity along a direction generating an edge (extreme ray) of $\text{rec}(P)$ — this is where $R(P)$ enters.

An application: Carathéodory's theorem. Minkowski's theorem represents each point of P as a combination of vertices and extreme rays, but gives no control on the *number* of terms used. The following classical result, stated here in its conic version, shows that a short representation always exists. Note that $y = Ax$ with $x \geq 0$ says precisely that y is a nonnegative combination of the columns of A , with coefficients x . Below, the *support* of a vector $x \in \mathbb{R}^k$ is the set of coordinates on which it is nonzero, denoted $\text{supp}(x) := \{j : x_j \neq 0\}$. Departing from our usual row convention, in this theorem and its proof subscripts on A select *columns*: A_j is the j -th column, and A_S the column submatrix indexed by S .

THEOREM 15 (CARATHÉODORY'S THEOREM, CONIC VERSION) *Let $A \in \mathbb{R}^{n \times k}$ and let $y = Ax$ for some $x \geq 0$. Then there exists $x' \geq 0$ with $y = Ax'$ whose support columns $\{A_j : j \in \text{supp}(x')\}$ are linearly independent; in particular, $|\text{supp}(x')| \leq \text{rank}(A)$.*

Exercise 2 Prove Theorem 15. (Hint: Show that $\{x : Ax = y, x \geq 0\}$ has vertices and that any vertex solution has the desired support bound.)

REMARK 3 (CARATHÉODORY REFINES MINKOWSKI) Applied to the decomposition of Theorem 13, Carathéodory's theorem bounds the number of generators needed for any single point: every $x \in P$ can be written as $x = \sum_i \lambda_i v_i + \sum_j \mu_j r_j$ with $v_i \in V(P)$, $r_j \in R(P)$, $\lambda, \mu \geq 0$, $\sum_i \lambda_i = 1$, using at most $\dim(P) + 1$ terms in total. To see this, lift to $\mathbb{R}^{n+1}$: the point $(x, 1)$ is a nonnegative combination of the vectors $\{(v, 1) : v \in V(P)\}$ and $\{(r, 0) : r \in R(P)\}$, all of which lie in the $(\dim(P) + 1)$ -dimensional subspace $\text{span}(\text{aff}(P) \times \{1\})$, and Theorem 15 reduces the support to at most $\dim(P) + 1$ columns. The unit last coordinate of $(x, 1)$ ensures the convex weights still sum to 1.

2.3 Solution Status of a Linear Program

With Theorem 13 in hand, classifying the possible outcomes of an LP takes only a few lines.

THEOREM 16 (TRICHOTOMY) *Consider $\max \{c^T x : Ax \leq b\}$ where $A \in \mathbb{R}^{m \times n}$ has full column rank. Then exactly one of the following holds:*

- (a) *the LP is infeasible: $P = \emptyset$;*
- (b) *the LP is feasible and unbounded: $\exists x \in P, r \in R(P)$ such that $c^T r > 0$: the feasible ray $x + tr$, $t \geq 0$, has $c^T(x + tr) \rightarrow \infty$;*
- (c) *the LP has an optimal vertex solution: $\text{OPT} = \max_{v \in V(P)} c^T v$.*

PROOF: Suppose $P \neq \emptyset$ and $c^T r \leq 0$ for all $r \in R(P)$; we prove (c). By Theorem 13, $V(P) \neq \emptyset$, and any $x \in P$ decomposes as $x = \sum_v \lambda_v v + \sum_r \mu_r r$, whence

$$c^T x = \sum_{v \in V(P)} \lambda_v c^T v + \sum_{r \in R(P)} \mu_r \underbrace{c^T r}_{\leq 0} \leq \sum_{v \in V(P)} \lambda_v c^T v \leq \max_{v \in V(P)} c^T v :$$

a weighted average is at most its best term. So the best vertex value bounds all of P and is attained — by that vertex. If instead some $r \in R(P)$ has $c^T r > 0$, then $r \in \text{rec}(P)$ makes the ray $x + tr$ feasible for every $x \in P$, with objective $c^T x + t c^T r \rightarrow \infty$: case (b). Mutual exclusivity is clear: (a) excludes the others, and under (b) the supremum is $+\infty$, while under (c) it is a finite maximum over the (finite, nonempty) vertex set. $\square$

COROLLARY 17 (A FIRST, INEFFICIENT, ALGORITHM) *An LP $\max \{c^\top x : \mathbf{A}x \leq b\}$ with $\text{rank}(\mathbf{A}) = n$ can be solved by exhaustive enumeration: for each $J \subseteq [m]$ with $|J| = n$ and $\mathbf{A}_J$ invertible, compute the candidate point $x_J := \mathbf{A}_J^{-1}b_J$ and test $\mathbf{A}x_J \leq b$; for each J with $|J| = n - 1$ and $\text{rank}(\mathbf{A}_J) = n - 1$, compute d_J spanning $\ker(\mathbf{A}_J)$ and record $\pm d_J$ as a candidate ray if $\mathbf{A}(\pm d_J) \leq \mathbf{0}$. If no candidate point is feasible, report infeasible; else, if some candidate ray r has $c^\top r > 0$, report unbounded; else, return the best feasible candidate point.*

PROOF: Every vertex is a candidate point by the proof of Corollary 7, and every $r \in R(P)$ is a positive multiple of a candidate ray: the tight set of its edge of $\text{rec}(P)$ has rank $n - 1$, and any $n - 1$ linearly independent rows J from it satisfy $\ker(\mathbf{A}_J) = \text{span}(r)$. A nonempty P has a vertex (as shown in the proof of Theorem 16), so feasibility is detected correctly. If a feasible candidate ray has $c^\top r > 0$ the LP is clearly unbounded; conversely, in case (b) of Theorem 16 the certifying $r \in R(P)$ appears among the candidate rays. Correctness of the final case is Theorem 16(c). $\square$

This brute-force algorithm is finite but impractical: $\binom{m}{n}$ grows exponentially. The simplex method (next lecture) can be seen as a far more efficient exploration of the same candidate set, walking only between *adjacent* feasible vertices while improving the objective.

3 Duality: Certifying Optimality

A remarkable feature of linear programming is that the solution status of an LP — and, in the feasible and bounded case, its precise optimal value — can be *succinctly certified*. We saw in Figure 2 that (3,1) appears to be optimal for the production LP, but a picture is not a proof. Duality turns the picture into a one-line arithmetic certificate.

3.1 Deriving inequalities, and the dual LP

The key idea is the following. From the system $\mathbf{A}x \leq b$ we can *derive* new valid inequalities: multiply each constraint $a_i^\top x \leq b_i$ by a multiplier $y_i \geq 0$ (nonnegativity keeps the inequality pointing the same way) and sum. Every feasible x then satisfies the derived inequality

$$(\mathbf{A}^\top y)^\top x = \sum_{i=1}^m y_i a_i^\top x \leq \sum_{i=1}^m y_i b_i = b^\top y.$$

If we can choose $y \geq \mathbf{0}$ so that the derived inequality has left hand side exactly $c^\top x$, that is $\mathbf{A}^\top y = c$, then $b^\top y$ is a certified upper bound on the optimal value. For the production LP (2), take $y = (\frac{3}{2}, \frac{1}{2}, 0, 0)$ for the constraints (wood, labor, $-x_1 \leq 0$, $-x_2 \leq 0$):

$$\frac{3}{2}(x_1 + x_2) + \frac{1}{2}(x_1 + 3x_2) = 2x_1 + 3x_2 \leq \frac{3}{2} \cdot 4 + \frac{1}{2} \cdot 6 = 9.$$

Together with the feasible point (3,1) of value 9, these two lines *prove* $\text{OPT} = 9$. Searching for the best such certificate is itself an LP, called the *dual*:

$$(\text{primal}) \quad \max \{c^\top x : \mathbf{A}x \leq b\} \qquad (\text{dual}) \quad \min \{b^\top y : \mathbf{A}^\top y = c, y \geq \mathbf{0}\}. \quad (3)$$

Mirroring the primal conventions, we declare the dual minimum to be $+\infty$ if the dual is infeasible and $-\infty$ if it is unbounded (from below). The computation above proves:

THEOREM 18 (WEAK DUALITY) For every primal feasible x and dual feasible y , $c^\top x \leq b^\top y$. Hence $\max \{c^\top x : \mathbf{A}x \leq b\} \leq \min \{b^\top y : \mathbf{A}^\top y = c, y \geq \mathbf{0}\}$.

PROOF: $c^\top x = (\mathbf{A}^\top y)^\top x = y^\top (\mathbf{A}x) \leq y^\top b$, using $\mathbf{A}^\top y = c$, then $y \geq \mathbf{0}$ and $\mathbf{A}x \leq b$. $\square$

Geometrically, a dual solution expresses the objective c as a nonnegative combination of constraint normals a_i . At an optimal vertex, the picture is particularly clean: c lies in the *cone* spanned by the normals of the tight constraints (Figure 6), which is exactly why the objective cannot be improved by moving in any feasible direction. This suggests — correctly — that optimal dual multipliers can be supported on tight constraints only (Exercise 3).

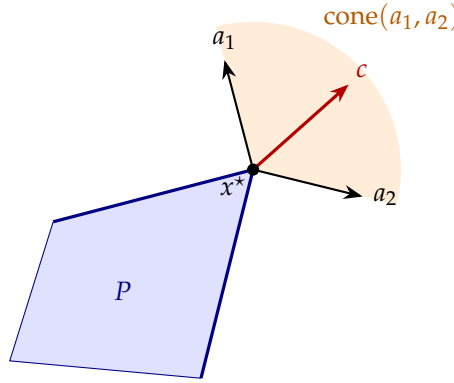


Figure 6: Duality, geometrically: at an optimal vertex x^* the objective c lies in the cone of the tight constraint normals, i.e., $c = y_1 a_1 + y_2 a_2$ with $y_1, y_2 \geq 0$. The multipliers (y_1, y_2) form an optimal dual solution.

The remarkable fact — *strong duality* — is that this certification scheme never loses anything: whenever the primal has a finite optimal value, a dual certificate of that exact value exists. Our proof will be constructive, based on the oldest algorithm of the subject: Fourier–Motzkin elimination.

3.2 Fourier–Motzkin elimination

Fourier–Motzkin (FM) elimination answers the following question: given a system $\mathbf{A}x \leq b$, what constraints does it impose on the variables $(x_1, \dots, x_{n-1})$ alone? Geometrically, we are asking for the *projection* of the polyhedron onto the first $n - 1$ coordinates — its “shadow” when the light shines along the x_n axis (Figure 7).

The procedure mirrors how one solves inequalities by hand. Split the constraints into three groups according to the sign of the coefficient of x_n . For the constraints where x_n appears with a non-zero coefficient, first scale the inequality by a positive number so that the coefficient in front of x_n becomes ± 1 , and then rearrange the inequality so that x_n appears alone on one side with coefficient $+1$. This rewrites the system as

$$\beta_i - g_i^\top x' \leq x_n, \quad i \in L, \quad x_n \leq \beta_j - g_j^\top x', \quad j \in U, \quad g_k^\top x' \leq \beta_k, \quad k \in Z,$$

where $x' = (x_1, \dots, x_{n-1})$: the constraints in U (L) give upper (lower) bounds on x_n , and those in Z do not mention x_n ; a constraint lands in L , U or Z according to whether the coefficient of x_n in

it is negative, positive or zero. A value of x_n compatible with x' exists if and only if every lower bound is at most every upper bound. This proves:

LEMMA 19 (FOURIER–MOTZKIN ELIMINATION [Fou27, Mot36]) *Given $x' \in \mathbb{R}^{n-1}$, there exists $x_n \in \mathbb{R}$ such that (x', x_n) satisfies the system above if and only if x' satisfies*

$$\beta_i - g_i^\top x' \leq \beta_j - g_j^\top x' \quad \forall (i, j) \in L \times U, \quad g_k^\top x' \leq \beta_k \quad \forall k \in Z.$$

Thus the projection of a polyhedron is again a polyhedron. Moreover, each derived constraint is a non-negative combination of the original constraints: scalings for Z , and sums of a scaled U -row and a scaled L -row for the pairs.

Let us make the “moreover” explicit, since it is the basis for all that follows. After the initial positive scaling, the i -th lower bound constraint reads

$$-x_n - g_i^\top x' \leq -\beta_i, \quad \text{and the } j\text{-th upper bound constraint reads} \quad x_n + g_j^\top x' \leq \beta_j,$$

both positive scalings of rows of the original system. Adding the two cancels x_n and yields $(g_j - g_i)^\top x' \leq \beta_j - \beta_i$, which is precisely the derived inequality $\beta_i - g_i^\top x' \leq \beta_j - g_j^\top x'$, rearranged. So every derived inequality is a nonnegative combination of at most two rows of the previous system, with weights the (positive) normalization factors.

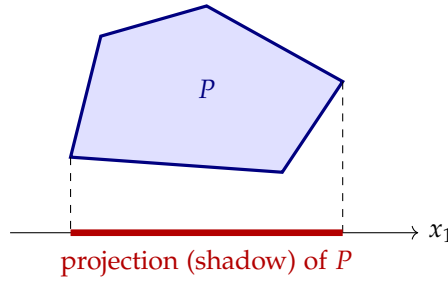


Figure 7: Eliminating x_2 computes the projection of P onto the x_1 -axis. By Lemma 19, the projection of a polyhedron is a polyhedron, described by nonnegative combinations of the original inequalities in which x_2 cancels.

Iterating Lemma 19, we can eliminate any number of variables. Since nonnegative combinations of nonnegative combinations are again nonnegative combinations, we obtain:

COROLLARY 20 *After eliminating any subset of the variables from $\mathbf{Ax} \leq b$, each derived inequality has the form $(\mathbf{A}^\top \mathbf{y})^\top x \leq b^\top \mathbf{y}$ for some $\mathbf{y} \geq \mathbf{0}$, where $\mathbf{A}^\top \mathbf{y}$ has zero coordinates on the eliminated variables. The solution set of the derived system is exactly the projection of $\{x : \mathbf{Ax} \leq b\}$ onto the remaining variables; in particular, it is solvable if and only if the original system is.*

Note that FM elimination is itself a (very inefficient) algorithm for solving LPs: the number of constraints can square with each eliminated variable, yielding $O(m^{2^n})$ in total. Its value for us is theoretical: it will enable us to show that every valid linear inequality for a polyhedron can be derived as a nonnegative combination of the constraints.

As an application, one can derive the converse of Minkowski’s theorem.

COROLLARY 21 (WEYL'S THEOREM [WEY35]) *For finite sets $V, R \subseteq \mathbb{R}^n$, the set $Q := \text{conv}(V) + \text{cone}(R)$ is a polyhedron. Combined with Theorem 13: a subset of $\mathbb{R}^n$ is a polyhedron if and only if it is finitely generated (the Minkowski–Weyl theorem).*

PROOF: Write $V = \{v^1, \dots, v^p\}$ and $R = \{r^1, \dots, r^q\}$. By definition, $x \in Q$ if and only if there exist $\lambda \in \mathbb{R}^p$ and $\mu \in \mathbb{R}^q$ with

$$x = \sum_{i=1}^p \lambda_i v^i + \sum_{j=1}^q \mu_j r^j, \quad \sum_{i=1}^p \lambda_i = 1, \quad \lambda \geq \mathbf{0}, \quad \mu \geq \mathbf{0}.$$

The set S of triples $(x, \lambda, \mu) \in \mathbb{R}^{n+p+q}$ satisfying these conditions is a polyhedron: each equality is the conjunction of two linear inequalities, and the sign conditions are linear inequalities. By construction, Q is the projection of S onto the x -coordinates, and by Corollary 20, eliminating the $p+q$ variables (λ, μ) one at a time yields a finite system of linear inequalities in x alone whose solution set is exactly this projection; hence Q is a polyhedron. $\square$

3.3 Strong duality

We say that the linear program $\max \{c^\top x : \mathbf{A}x \leq b\}$ is *bounded* if there exists a number $d \in \mathbb{R}$ such that $c^\top x \leq d$ for every feasible x ; similarly, a minimization problem is bounded if some $d \in \mathbb{R}$ satisfies $c^\top x \geq d$ for every feasible x .

THEOREM 22 (STRONG DUALITY [GKT51]) *Suppose $\max \{c^\top x : \mathbf{A}x \leq b\}$ is feasible and bounded. Then the primal and the dual $\min \{b^\top y : \mathbf{A}^\top y = c, y \geq \mathbf{0}\}$ both have optimal solutions, x^* and y^* respectively, and these satisfy $c^\top x^* = b^\top y^*$.*

PROOF: Consider the following system in the variables $(x, t) \in \mathbb{R}^n \times \mathbb{R}$:

$$t - c^\top x \leq 0, \quad \mathbf{A}x \leq b.$$

Its solutions are the pairs where x is primal feasible and $t \leq c^\top x$. Let $T \subseteq \mathbb{R}$ denote the projection of the solution set onto the t -coordinate:

$$T = \{t \in \mathbb{R} : t \leq c^\top x \text{ for some } x \in P\},$$

a nonempty set, as $P \neq \emptyset$.

Now eliminate the variables $x_1, \dots, x_n$ via FM. By Corollary 20, T is the solution set of a finite system of inequalities in t alone, each a nonnegative combination of the original rows: multiplier $\mu_k \geq 0$ on the row $t - c^\top x \leq 0$ and multipliers $y^k \geq \mathbf{0}$ on $\mathbf{A}x \leq b$, where cancellation of the x -coordinates means $\mathbf{A}^\top y^k = \mu_k c$, and the k -th derived inequality reads $\mu_k t \leq b^\top y^k$. A derived inequality with $\mu_k = 0$ does not involve t at all, and holds automatically since $T \neq \emptyset$; we discard it. Since T is bounded above — every $t \in T$ satisfies $t \leq c^\top x \leq d$ for some feasible x , by boundedness of the primal — at least one inequality with $\mu_k > 0$ remains. Dividing each remaining inequality by its μ_k — replacing (μ_k, y^k) by $(1, y^k/\mu_k)$, still a nonnegative combination of the original rows — we arrive at the description

$$T = \{t \in \mathbb{R} : t \leq b^\top y^k, k \in [N]\} = (-\infty, \min_{k \in [N]} b^\top y^k], \quad \text{where } \mathbf{A}^\top y^k = c, \quad y^k \geq \mathbf{0}.$$

Set $\tau := \min_{k \in [N]} b^\top y^k$. On one hand $\tau \in T$, so some feasible x^* has $c^\top x^* \geq \tau$; on the other hand $c^\top x \in T$ for every feasible x , so $c^\top x \leq \tau$. Hence x^* is primal optimal with $c^\top x^* = \tau$, and the minimizing $y^* := y^k$ is dual feasible with $b^\top y^* = \tau = c^\top x^*$. By weak duality (Theorem 18) no dual solution can have smaller value, so y^* is dual optimal. $\square$

Observe that the boundedness hypothesis is automatic when both programs are feasible: by weak duality, any dual feasible y certifies the bound $d = b^\top y$. Note also the complementary roles of the two duality theorems: weak duality is elementary but useful (any dual feasible point gives a bound), while strong duality guarantees that no gap remains — the *best* derived inequality is exactly tight.

Exercise 3 (Complementary slackness) Let x be primal feasible and y be dual feasible for (3). Show that x and y are both optimal if and only if $y_i(b_i - a_i^\top x) = 0$ for all $i \in [m]$, i.e., positive multipliers appear only on tight constraints. (Hint: Inspect where the proof of weak duality can be loose.)

3.4 Duality for general systems: a recipe

Linear systems rarely arrive in the tidy form $Ax \leq b$: in practice one faces a mix of $\leq$ -inequalities, $\geq$ -inequalities and equations,

$$A^u x \leq b^u, \quad A^\ell x \geq b^\ell, \quad A^e x = b^e. \quad (4)$$

One could convert (4) into the form (3) each time duality is needed, but the conversion is tedious and error-prone to redo. It is better to record, once and for all, a recipe for deriving valid inequalities directly from the general form. Suppose the goal is to prove that $c^\top x \leq d$ holds for every solution of (4).

1. *Orient.* Rearrange every inequality so that it points in the same direction as the target: multiply the $\geq$ -rows by -1 , turning $A^\ell x \geq b^\ell$ into $-A^\ell x \leq -b^\ell$. Equations need no orientation, since they may be used in both directions.
2. *Combine.* Scale the oriented inequalities by *nonnegative* multipliers $y^u, y^\ell \geq 0$ and the equations by *unrestricted* multipliers z , and add. Every solution x of (4) satisfies the resulting inequality

$$((A^u)^\top y^u - (A^\ell)^\top y^\ell + (A^e)^\top z)^\top x \leq (b^u)^\top y^u - (b^\ell)^\top y^\ell + (b^e)^\top z.$$

3. *Match.* Choose the multipliers so that the coefficient vector on the left equals c and the right-hand side is at most d . The derived inequality then implies $c^\top x \leq d$, certifying the target.

To certify an inequality of the form $c^\top x \geq d$ instead, orient the other way — now every inequality is rearranged so that x sits on the $\geq$ side — and again combine with nonnegative multipliers; this is the same as applying the recipe to $-c^\top x \leq -d$. Strong duality guarantees that the recipe is *complete*: whenever the target inequality is valid and the system is feasible, matching multipliers exist.

Minimizing the certified upper bound over all matching multipliers is precisely the *dual program* of maximizing $c^\top x$ over (4); dually, for the problem of *minimizing* $c^\top x$, one orients the

recipe as $\geq$ and *maximizes* the certified lower bound. The following theorem records the resulting primal–dual pairs for both orientations of the objective, as a “cheat sheet” to be consulted whenever a dual must be written down.

THEOREM 23 (DUALITY CHEAT SHEET) Let $P := \{x \in \mathbb{R}^n : \mathbf{A}^u x \leq b^u, \mathbf{A}^\ell x \geq b^\ell, \mathbf{A}^e x = b^e\}$, and index the multipliers y^u, y^ℓ and z by the $\leq, \geq$ - and $=$ -constraints respectively, with z unrestricted in sign. The following are primal–dual pairs:

$$\begin{array}{ll} \text{(P)} & \max c^\top x \\ & \text{s.t. } x \in P \\ \text{(D)} & \min (b^u)^\top y^u - (b^\ell)^\top y^\ell + (b^e)^\top z \\ & \text{s.t. } (\mathbf{A}^u)^\top y^u - (\mathbf{A}^\ell)^\top y^\ell + (\mathbf{A}^e)^\top z = c, \quad y^u, y^\ell \geq \mathbf{0}, \\ \text{(P')} & \min c^\top x \\ & \text{s.t. } x \in P \\ \text{(D')} & \max (b^\ell)^\top y^\ell - (b^u)^\top y^u + (b^e)^\top z \\ & \text{s.t. } (\mathbf{A}^\ell)^\top y^\ell - (\mathbf{A}^u)^\top y^u + (\mathbf{A}^e)^\top z = c, \quad y^u, y^\ell \geq \mathbf{0}. \end{array}$$

Weak duality holds for each pair: for all feasible solutions, the objective of (D) is an upper bound on that of (P), and the objective of (D') a lower bound on that of (P'). Strong duality holds as well: if the primal member of either pair is feasible and bounded, then both members have optimal solutions, and their objective values are equal.

We do not prove Theorem 23: every claim is essentially a syntactic manipulation of the duality theorems already established for the pair (3). As a mnemonic, each dual searches for the best bound derivable by the recipe: orient the inequalities in the direction being certified — $\leq$ when upper bounding a maximum, $\geq$ when lower bounding a minimum — combine them with nonnegative multipliers, use the equations freely, and optimize the resulting bound in the direction opposite to the primal.

REMARK 4 (WHY FEASIBILITY IS NEEDED) Feasibility is essential to the completeness of the recipe, and hence to the strong duality statement of Theorem 23. Over an infeasible system every inequality is vacuously valid, but need not be derivable: in the system $x_1 \leq 0, x_1 \geq 1$ in the variables (x_1, x_2) , the valid inequality $x_2 \leq 0$ can never be matched, as every combination of the rows has coefficient 0 on x_2 . The remedy is a certificate of infeasibility, such as the derivable contradiction $0 \leq -1$ — the subject of the next subsection.

3.5 The Farkas lemma

Strong duality certifies optimal values; its close relative, the Farkas lemma, certifies *infeasibility*. How can one demonstrate that a system $\mathbf{A}x \leq b$ has no solution? By exhibiting a contradiction derivable from the system: a nonnegative combination of the constraints that reads $0 \leq -1$.

THEOREM 24 (FARKAS LEMMA [FAR02]) The system $\mathbf{A}x \leq b$ is infeasible if and only if there exists $y \geq \mathbf{0}$ with $\mathbf{A}^\top y = \mathbf{0}$ and $b^\top y = -1$.

The proof is an instructive application of strong duality to an auxiliary LP that measures constraint violation; we leave it as an exercise.

Exercise 4 (Proof of the Farkas lemma) Prove Theorem 24. (Hint: $(\Leftarrow)$ Evaluate $y^\top \mathbf{A}x$ in two ways. $(\Rightarrow)$ Apply strong duality to the “phase one” LP $\min \{\mathbf{1}^\top z : \mathbf{A}x - z \leq b, z \geq \mathbf{0}\}$, which measures the minimum total constraint violation, and normalize an optimal dual solution.)

The Farkas lemma has a remarkable consequence: there always exists a “short proof” to demonstrate the feasibility status of a linear system $\mathbf{Ax} \leq \mathbf{b}$. To prove feasibility, one needs only produce a valid solution x , which can be checked by substitution. For infeasibility, the proof is the nonnegative multiplier vector y satisfying $\mathbf{A}^\top y = \mathbf{0}$, $\mathbf{b}^\top y = -1$, whose validity is again checked by simple substitution. Thus, whatever the status, it admits a short and easily verifiable certificate.

For further reading on today’s material, we recommend Matoušek and Gärtner [MG07] for a gentle treatment and Schrijver [Sch86] for the definitive reference.

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